

Finally `model.summary()` prints a summary of the U-Net model structure, including each layer in each block, the layer type, the output shape of each layer, the number of parameters in each layer, and the layer's connection to other layers. Please note, that the output shape always starts with `None`. This is the batch size of the data, and this is not fixed nor set in our model.

More details on the U-Net model and it's structure can be found in `UNET.ipynb` or [here](#)

```
In [12]: # Build the U-Net
# Specify the details of the optimizer
optimizer = optimizers.Adam(learning_rate=1e-4) # Adam optimizer with a learning r

# Create the model
# NOTE: We have chosen to use some custom metrics iou_metric and dice_coeff. The de
metrics = [iou_metric, dice_coeff] # List of custom metrics to use
custom_objects = {"iou_metric": iou_metric, "dice_coeff": dice_coeff} # Custom obj

model = unet_model(
    the_weights=[0.26, 3.69/8, 3.31/8, 2.56/8, 6.67/8],
    input_shape=(256, 256, 3), # Input shape of the images
    num_classes=num_classes,   # Number of classes
    num_blocks=4,              # Number of blocks in the U-Net model
    optimizer=optimizer,       # Optimizer to use
    metrics=metrics            # Metrics to use
)

# Get a print out of the model structure.
model.summary() # Print the summary of the model
```

Weights: [0.26, 0.46125, 0.41375, 0.32, 0.83375]

Model: "functional"

4/4  6s 334ms/step

Metrics based on test data:

Class Background Individual Metrics:

Precision: 0.9046585846497925

Recall: 0.9634046695098679

F1 Score: 0.9331079157229737

Class Low to Moderate Risk Individual Metrics:

Precision: 0.6458637340946741

Recall: 0.5656960446535522

F1 Score: 0.6031275817690487

Class High Risk Individual Metrics:

Precision: 0.6939090122354028

Recall: 0.5342760423362043

F1 Score: 0.6037184045166062

Class Very High Risk Individual Metrics:

Precision: 0.7362773340155818

Recall: 0.5637340676602934

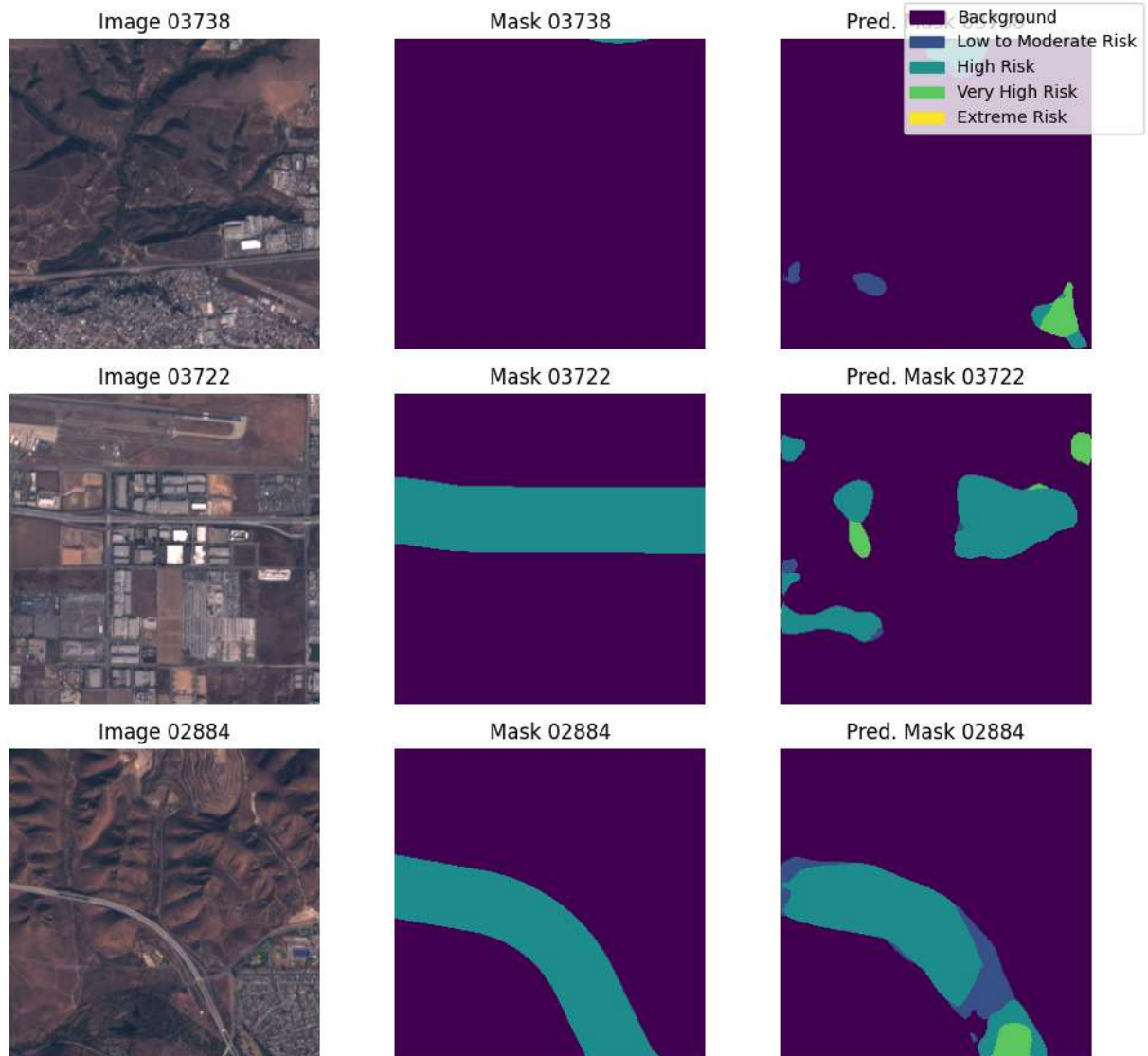
F1 Score: 0.6385553478925045

Class Extreme Risk Individual Metrics:

Precision: 0.8545546201542572

Recall: 0.5406858805067555

F1 Score: 0.66231680777661



Prediction Results

From the example run, we can see that the model performed quite well in predicting the threat risks. **These results demonstrate the practical value and real-world applicability of a U-Net model for segmenting wildfire threats near critical transportation infrastructure.** With high F1-scores across key threat levels, especially very high (0.81) and extreme (0.85), **the model shows that it can detect the most hazardous areas, making it an effective tool for emergency planning and infrastructure risk assessment.** The background class achieved an F1-score of 0.96, indicating the model is highly effective at distinguishing fire-affected areas from non-threat regions.

Additional performance across remaining threat levels also shows meaningful predictive power. Despite slightly lower recall scores for High Risk and Low-to-Medium Risk, the model still achieved F1-scores of 0.73 and 0.76 respectively. **This indicates reasonable performance for more subtle or less spatially distinct fire zones.**